

SQL_TASK3_PROJECT

Readme.md

RESULTS.md

Sales_Analysis_Project.ipynb

sales_analysis.db

Sample - Superstore.csv

Sales_Analysis_Project.ipynb

df.to_sql()

GenerateCodeMarkdownRun AllRestartClear All OutputsPython 3.14.2

```
pd.read_sql(
    "SELECT * FROM superstore_raw LIMIT 5",
    conn
)
```

[11] ✓ 0.0s Python

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	Pos Co
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	424
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	424
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	900
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	33
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	33

5 rows × 21 columns

CUSTOMER TABLE

conn.execute("""

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df.to_sql()

Python 3.14.2

CHAT

• CUSTOMER TABLE

```
conn.execute("""
CREATE TABLE customers AS
SELECT DISTINCT
  "Customer ID" AS customer_id,
  "Customer Name" AS customer_name,
  Segment,
  Country,
  City,
  state,
  Region
FROM superstore_raw;
""")
```

[14] ✓ 0.0s Python

<sqlite3.Cursor at 0x1107680c0>

```
pd.read_sql(
  "SELECT * FROM customers LIMIT 5",
  conn
)
```

[15] ✓ 0.0s Python

	customer_id	customer_name	Segment	Country	City	State	Region
0	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	South
1	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	West
2	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	South
3	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	West
4	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	South

• PRODUCT TABLE

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PRODUCT TABLE

```
conn.execute("""
CREATE TABLE products AS
SELECT DISTINCT
    "Product ID" AS product_id,
    "Product Name" AS product_name,
    Category,
    "Sub-Category" AS sub_category
FROM superstore_raw;
""")
```

[16] ✓ 0.0s

<sqlite3.Cursor at 0x110777540>

```
pd.read_sql(
    "SELECT * FROM products LIMIT 5",conn)
```

[18] ✓ 0.0s

	product_id	product_name	Category	sub_category
0	FUR-BO-10001798	Bush Somerset Collection Bookcase	Furniture	Bookcases
1	FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs,...	Furniture	Chairs
2	OFF-LA-10000240	Self-Adhesive Address Labels for Typewriters b...	Office Supplies	Labels
3	FUR-TA-10000577	Bretford CR4500 Series Slim Rectangular Table	Furniture	Tables
4	OFF-ST-10000760	Eldon Fold 'N Roll Cart System	Office Supplies	Storage

ORDER TABLE

```
conn.execute("""
CREATE TABLE orders AS
SELECT
    "Order ID" AS order_id,
    "Order Date" AS order_date,
```

[19] ✓ 0.0s

Select Cell Language Mode

Python

Python

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ORDER TABLE

```
conn.execute("""
CREATE TABLE orders AS
SELECT
    "Order ID" AS order_id,
    "Order Date" AS order_date,
    "Ship Date" AS ship_date,
    "Ship Mode" AS ship_mode,
    "Customer ID" AS customer_id,
    "Product ID" AS product_id,
    Sales,
    Quantity,
    Discount,
    Profit
FROM superstore_raw;
""")
```

[19] ✓ 0.0sPython

<sqlite3.Cursor at 0x1107761c0>

```
pd.read_sql(""" \
"SELECT * FROM orders LIMIT 5",conn)
```

[20] ✓ 0.0sPython

	order_id	order_date	ship_date	ship_mode	customer_id	product_id	Sales	Quantity	Discount	Proi
0	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-BO-10001798	261.9600	2	0.00	41.913
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-CH-10000454	731.9400	3	0.00	219.582
2	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	OFF-LA-10000240	14.6200	2	0.00	6.87
3	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45	-383.03
	US-					OFF-PT				

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• verify tables

```
pd.read_sql(
    "SELECT name FROM sqlite_master WHERE type='table';",
    conn
)
```

[21] ✓ 0.0sPython

	name
0	superstore_raw
1	customers
2	products
3	orders

Step 2: Perform Required Queries

Write and execute SQL queries for each of the following:

1.Find all orders where sales are greater than the average sales. (Subquery)

2.Find the highest sales order for each customer. (Subquery)

3.Calculate total sales for each customer. (CTE)

4.Find customers whose total sales are above average. (CTE + Subquery)

5.Rank all customers based on total sales. (Window Function)

6.Assign row numbers to each order within a customer. (Window Function + PARTITION BY)

7.Display top 3 customers based on total sales. (Window Function)

1.) Find all orders where sales are greater than the average sales. (Subquery)

```
pd.read_sql(
```

[23] ✓ 0.0sPython

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1.) Find all orders where sales are greater than the average sales. (Subquery)

```
pd.read_sql(
    """SELECT *
    FROM orders
    WHERE sales>(
        SELECT AVG(Sales)
        FROM orders );
    """, conn)
```

[23] ✓ 0.0sPython

	order_id	order_date	ship_date	ship_mode	customer_id	product_id	Sales	Quantity	Discount		
...	0	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-BQ-10001798	261.9600	2	0.00	4
	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-CH-10000454	731.9400	3	0.00	21
	2	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45	-38
	3	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002275	907.1520	6	0.20	9
	4	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	FUR-TA-10001539	1706.1840	9	0.20	8
...	...	...	...	...	...	...	...	...	...	...	...
	2355	US-2016-103674	12/6/2016	12/10/2016	Standard Class	AP-10720	TEC-PH-10004080	271.9600	5	0.20	2
	2356	US-2016-103674	12/6/2016	12/10/2016	Standard Class	AP-10720	TEC-PH-10002496	249.5840	2	0.20	3
	2357	US-2016-103674	12/6/2016	12/10/2016	Standard Class	AP-10720	OFF-BI-10002026	437.4720	14	0.20	15
	2358	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	DB-13060	TEC-PH-10003645	258.5760	2	0.20	1
	2359	CA-2017-	5/4/2017	5/9/2017	Second	CG-12220	OFF-AP-	243.1600	2	0.00	7

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2. Find the highest sales order for each customer (Subquery)

```
pd.read_sql("""
SELECT customer_id, order_id,Sales
FROM orders o
WHERE sales=
(
SELECT MAX(SALES)
FROM orders
WHERE customer_id=o.customer_id
);
""",conn)
```

[35] ✓ 8.8s

Python

	customer_id	order_id	Sales
0	CG-12520	CA-2016-162156	731.9400
1	SO-20336	US-2015-108966	957.5775
2	BH-11710	CA-2014-115812	1706.1840
3	EB-13870	CA-2015-106320	1044.6300
4	TB-21520	US-2015-150630	3083.4300
...	...	...	...
790	DH-13075	CA-2015-159534	1087.9360
791	IM-15055	CA-2016-129630	2799.9600
792	HW-14935	CA-2017-121559	2405.2000
793	RB-19435	CA-2017-163871	735.9800
794	AP-10720	US-2016-103674	437.4720

795 rows x 3 columns

3. Calculate total sales for each customer (CTE)

```
pd.read_sql("""
WITH customer_sales AS
/SQL FCT
```

[37] ✓ 0.0s

Python

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3. Calculate total sales for each customer (CTE)

```
pd.read_sql("""
WITH customer_sales AS
(SELECT
    customer_id,
    SUM(Sales) AS total_sales
FROM orders
GROUP BY customer_id
)

SELECT *
FROM customer_sales
ORDER BY total_sales DESC; """, conn)
```

[37] ✓ 0.0s Python

	customer_id	total_sales
0	SM-20320	25043.050
1	TC-20980	19052.218
2	RB-19360	15117.339
3	TA-21385	14595.620
4	AB-10105	14473.571
...	...	...
788	RS-19870	22.328
789	MG-18205	16.739
790	CJ-11875	16.520
791	LD-16855	5.304
792	TS-21085	4.833

793 rows x 2 columns

4. Find customers whose total sales are above average (CTE + Subquery)

```
pd.read_sql("""
SELECT *
FROM customer_sales
WHERE total_sales > (SELECT AVG(total_sales) FROM customer_sales); """, conn)
```

[38] ✓ 0.0s Python

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4. Find customers whose total sales are above average (CTE + Subquery)

```
pd.read_sql("""
WITH customer_sales AS
(
    SELECT customer_id,
    SUM(Sales) AS total_sales
    FROM orders
    GROUP BY customer_id
)
SELECT *
FROM customer_sales
WHERE total_sales>
(
    SELECT AVG(total_sales)
    FROM customer_sales
);
""",conn)
```

[38] ✓ 0.0s Python

	customer_id	total_sales
0	AA-10315	5583.560
1	AA-10645	5086.935
2	AB-10060	7755.620
3	AB-10105	14473.571
4	AC-10450	5527.846
...	...	...
289	VW-21775	6134.038
290	WB-21850	6160.102
291	YC-21895	5454.350
292	YS-21880	6720.444
293	ZC-21910	8025.707

294 rows x 2 columns

5. Rank all customers based on total sales (Window Function)

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5. Rank all customers based on total sales (Window Function)

```
pd.read_sql("""
WITH customer_sales AS
(
    SELECT customer_id , SUM(Sales) AS total_sales
    FROM orders
    GROUP BY customer_id
)
SELECT customer_id , total_sales,
RANK() OVER( ORDER BY total_sales DESC) AS customer_rnk
FROM customer_sales;""",conn)
```

[39] ✓ 0.0s Python

	customer_id	total_sales	customer_rnk
0	SM-20320	25043.050	1
1	TC-20980	19052.218	2
2	RB-19360	15117.339	3
3	TA-21385	14595.620	4
4	AB-10105	14473.571	5
...	...	...	...
788	RS-19870	22.328	789
789	MG-18205	16.739	790
790	CJ-11875	16.520	791
791	LD-16855	5.304	792
792	TS-21085	4.833	793

793 rows x 3 columns

6. Assign row numbers to each order within a customer (PARTITION BY)

```
pd.read_sql("""
SELECT customer_id , order_id , Sales
```

[42] ✓ 0.6s Python

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6. Assign row numbers to each order within a customer (PARTITION BY)

```
pd.read_sql("""
SELECT customer_id , order_id, Sales,
ROW_NUMBER() OVER(
PARTITION BY customer_id
ORDER BY Sales DESC) AS row_num
FROM orders;
""",conn)
```

[42] ✓ 0.6s Python

	customer_id	order_id	Sales	row_num
0	AA-10315	CA-2016-103982	3930.072	1
1	AA-10315	CA-2014-128055	673.568	2
2	AA-10315	CA-2016-103982	431.976	3
3	AA-10315	CA-2017-147039	362.940	4
4	AA-10315	CA-2014-128055	52.080	5
...	...	...	...	...
9989	ZD-21925	CA-2017-141481	61.440	5
9990	ZD-21925	CA-2014-143336	22.720	6
9991	ZD-21925	US-2016-147991	16.720	7
9992	ZD-21925	CA-2016-152471	15.984	8
9993	ZD-21925	CA-2014-143336	8.560	9

9994 rows x 4 columns

7. Display Top 3 Customers Based on Total Sales (Window Function)

```
pd.read_sql("""
WITH customer_sales AS
(
SELECT customer_id, SUM(Sales) AS total_sales
FROM orders

```

[43] ✓ 0.0s Python

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7. Display Top 3 Customers Based on Total Sales (Window Function)

```
pd.read_sql("""
WITH customer_sales AS
(
    SELECT customer_id, SUM(Sales) AS total_sales
    FROM orders
    GROUP BY customer_id
),
ranked_customers AS
(
    SELECT customer_id, total_sales,
    RANK() OVER(
        ORDER BY total_sales DESC) AS sales_rank
    FROM customer_sales
)
SELECT *
FROM ranked_customers
WHERE sales_rank<=3;

""", conn)
```

[43] ✓ 0.0s Python

	customer_id	total_sales	sales_rank
0	SM-20320	25043.050	1
1	TC-20980	19052.218	2
2	RB-19360	15117.339	3

Step 3: Final Combined Query

Write one final query that shows:

Customer Name

Total Sales

Rank

(Use JOIN + CTE + Window Function together)

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7. Display Top 3 Customers Based on Total Sales (Window Function)

```
pd.read_sql("""
WITH customer_sales AS
(
    SELECT customer_id, SUM(Sales) AS total_sales
    FROM orders
    GROUP BY customer_id
),
ranked_customers AS
(
    SELECT customer_id, total_sales,
    RANK() OVER(
        ORDER BY total_sales DESC) AS sales_rank
    FROM customer_sales
)
SELECT *
FROM ranked_customers
WHERE sales_rank<=3;

""", conn)
```

[43] ✓ 0.0s Python

	customer_id	total_sales	sales_rank
0	SM-20320	25043.050	1
1	TC-20980	19052.218	2
2	RB-19360	15117.339	3

Step 3: Final Combined Query

Write one final query that shows:

Customer Name

Total Sales

Rank

(Use JOIN + CTE + Window Function together)

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Step 3: Final Combined Query

Write one final query that shows:

Customer Name

Total Sales

Rank

(Use JOIN + CTE + Window Function together)

[44] ✓ 0.1s

	name
0	superstore_raw
1	customers
2	products
3	orders

[45] ✓ 0.0s

	cid	name	type	notnull	dfbt_value	pk
0	0	customer_id	TEXT	0	None	0
1	1	customer_name	TEXT	0	None	0
2	2	Segment	TEXT	0	None	0
3	3	Country	TEXT	0	None	0
4	4	City	TEXT	0	None	0
5	5	State	TEXT	0	None	0

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• FINAL COMBINED QUERY

```
final_result= pd.read_sql("""
WITH customer_sales AS
(
    SELECT c.customer_id, c.customer_name, SUM(o.Sales) AS total_sales
    FROM customers c
    JOIN orders o
    ON c.customer_id=o.customer_id
    GROUP BY c.customer_id, c.customer_name
)
SELECT customer_name, total_sales,
       RANK() OVER(
           ORDER BY total_sales DESC ) AS customer_rank
FROM customer_sales;
""",conn)
final_result.head(20)
```

[54] ✓ 0.2s Python

	customer_name	total_sales	customer_rank
0	Ken Lonsdale	141752.290	1
1	Sanjit Engle	134303.818	2
2	Adrian Barton	130262.139	3
3	Sean Miller	125215.250	4
4	Clay Ludtke	119686.006	5
5	Sanjit Chand	113138.672	6
6	Seth Vernon	103238.550	7
7	Zuschuss Carroll	96308.484	8
8	Maria Etezadi	95973.552	9
9	Laura Armstrong	95405.442	10
10	Tamara Chand	95261.090	11
11	Anna Häberlin	94659.528	12
12	Greg Tran	94560.960	13
13	Edward Hooks	92797.920	14
14	Raymond Buch	90704.034	15
15	Caroline Jumper	89319.792	16

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SUGGESTED ACTIONS

Build Workspace

Show Config

Sales_Analysis_Project.ipynb • Cell 9 +

Describe what to build next

Agent Auto

Spaces: 4

Finish Setup

Cell 9 of 54

SQL_TASK3_PROJECT

Readme.md

RESULTS.md

Sales_Analysis_Project.ipynb

sales_analysis.db

Sample - Superstore.csv

Sales_Analysis_Project.ipynb — sql_task3_project

Readme.mdSales_Analysis_Project.ipynbRESULTS.mdSample - Superstore.csv

df.to_sql()

GenerateCodeMarkdownRun AllRestartClear All OutputsPython 3.14.2

FINAL COMBINED QUERY

```
final_result= pd.read_sql("""
WITH customer_sales AS
(
    SELECT c.customer_id, c.customer_name, SUM(o.Sales) AS total_sales
    FROM customers c
    JOIN orders o
    ON c.customer_id=o.customer_id
    GROUP BY c.customer_id, c.customer_name
)
SELECT customer_name, total_sales,
       RANK() OVER(
           ORDER BY total_sales DESC ) AS customer_rank
FROM customer_sales;
""",conn)
final_result.head(20)
```

[54] ✓ 0.2sPython

	customer_name	total_sales	customer_rank
0	Ken Lonsdale	141752.290	1
1	Sanjit Engle	134303.818	2
2	Adrian Barton	130262.139	3
3	Sean Miller	125215.250	4
4	Clay Ludtke	119686.006	5
5	Sanjit Chand	113138.672	6
6	Seth Vernon	103238.550	7
7	Zuschuss Carroll	96308.484	8
8	Maria Etezadi	95973.552	9
9	Laura Armstrong	95405.442	10
10	Tamara Chand	95261.090	11
11	Anna Häberlin	94659.528	12
12	Greg Tran	94560.960	13
13	Edward Hooks	92797.920	14
14	Raymond Buch	90704.034	15
15	Caroline Jumper	89319.792	16

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SUGGESTED ACTIONS

Build WorkspaceShow Config

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Describe what to build next

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SQL_TASK3_PROJECT

Readme.md

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sales_analysis.db

Sample - Superstore.csv

Sales_Analysis_Project.ipynb > df.to_sql()

Generate + Code + Markdown | Run All Restart Clear All Outputs ...

Python 3.14.2

1.)Who are the top 5 customers? Insight:

The top 5 customers contribute the highest revenue and represent the most valuable customer segment.

```
pd.read_sql("""
WITH customer_sales AS
(
    SELECT c.customer_id, c.customer_name, SUM(o.Sales) AS total_sales
    FROM customers c
    JOIN orders o
    ON c.customer_id = o.customer_id
    GROUP BY c.customer_id, c.customer_name
)
SELECT *
FROM customer_sales
ORDER BY total_sales DESC
LIMIT 5;
""",conn)
```

0.1s

Python

	customer_id	customer_name	total_sales
0	KL-16645	Ken Lonsdale	141752.290
1	SE-20110	Sanjit Singh	134303.818
2	AB-10105	Adrian Barton	130262.139
3	SM-20320	Sean Miller	125215.250
4	CL-12565	Clay Ludtke	119686.006

2.)Who are the bottom 5 customers?

markdown

```
pd.read_sql("""
WITH customer_sales AS
(
    SELECT c.customer_id, c.customer_name, SUM(o.Sales) AS total_sales
    FROM customers c
    JOIN orders o
    ON c.customer_id = o.customer_id
    GROUP BY c.customer_id, c.customer_name
)
SELECT *
FROM customer_sales
ORDER BY total_sales ASC
LIMIT 5;
""",conn)
```

0.3s

Python

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SUGGESTED ACTIONS

Build Workspace Show Config

Sales_Analysis_F

Describe what to

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SQL_TASK3_PROJECT

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Sample - Superstore.csv

Sales_Analysis_Project.ipynb — sql_task3_project

Generate + Code + Markdown | Run All Restart Clear All Outputs Python 3.14.2

2.) Who are the bottom 5 customers?

markdown

```
pd.read_sql("""
WITH customer_sales AS
(
    SELECT c.customer_id, c.customer_name, SUM(o.Sales) AS total_sales
    FROM customers c
    JOIN orders o
    ON c.customer_id=o.customer_id
    GROUP BY c.customer_id, c.customer_name
)
SELECT *
FROM customer_sales
ORDER BY total_sales ASC
LIMIT 5;
""", conn)
```

Python

[56] ✓ 0.3s

	customer_id	customer_name	total_sales
0	LD-16855	Lela Donovan	5.304
1	TS-21085	Thais Sissman	9.666
2	CJ-11875	Carl Jackson	16.520
3	MG-18205	Mitch Gastineau	16.739
4	RS-19870	Roy Skaria	44.656

3.) Which customers made only one order? Insight:

These customers may be targeted through retention campaigns to encourage repeat purchases.

```
pd.read_sql("""SELECT c.customer_id, c.customer_name, COUNT(o.order_id) AS total_orders
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
```

Python

[62] ✓ 0.3s

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SUGGESTED ACTIONS

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Sales_Analysis_F

Describe what to t

Agent Auto

SQL_TASK3_PROJECT

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RESULTS.md

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sales_analysis.db

Sample - Superstore.csv

Sales_Analysis_Project.ipynb

df.to_sql()

Generate + Code + Markdown | Run All Restart Clear All Outputs

Python 3.14.2

3.)Which customers made only one order? Insight:

These customers may be targeted through retention campaigns to encourage repeat purchases.

```
pd.read_sql("""SELECT c.customer_id, c.customer_name, COUNT(o.order_id) AS total_orders
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name
HAVING COUNT(o.order_id)=1;
""", conn)
```

[62] ✓ 0.3s Python

	customer_id	customer_name	total_orders
0	AO-10810	Anthony O'Donnell	1
1	CJ-11875	Carl Jackson	1
2	JR-15700	Jocasta Rupert	1
3	LD-16855	Lela Donovan	1
4	RE-19405	Ricardo Emerson	1

4.)Which customers have above-average sales? Insight:

Customers above the average sales threshold are strong candidates for loyalty and premium programs.

```
pd.read_sql("""
WITH customer_sales AS
(
SELECT c.customer_id , c.customer_name,SUM(o.Sales) AS total_sales
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name
)
SELECT *
FROM customer_sales
WHERE total sales>
```

[63] ✓ 0.3s Python

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SUGGESTED ACTIONS

Build Workspace Show Config

Sales_Analysis_F

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SQL_TASK3_PROJECT

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Sales_Analysis_Project.ipynb > df.to_sql()

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4.)Which customers have above-average sales? Insight:

Customers above the average sales threshold are strong candidates for loyalty and premium programs.

```
pd.read_sql("""
WITH customer_sales AS
(
    SELECT c.customer_id , c.customer_name,SUM(o.Sales) AS total_sales
    FROM customers c
    JOIN orders o
    ON c.customer_id = o.customer_id
    GROUP BY c.customer_id, c.customer_name
)
SELECT *
FROM customer_sales
WHERE total_sales>
(
    SELECT AVG(total_sales)
    FROM customer_sales
)
ORDER BY total_sales DESC;
""",conn)
```

[63] ✓ 0.3s Python

	customer_id	customer_name	total_sales
0	KL-16645	Ken Lonsdale	141752.290
1	SE-20110	Sanjit Engle	134303.818
2	AB-10105	Adrian Barton	130262.139
3	SM-20320	Sean Miller	125215.250
4	CL-12565	Clay Ludtke	119686.006
...	...	...	...
273	TS-21610	Troy Staebel	19742.968
274	BD-11605	Brian Dahlen	19730.826
275	FM-14215	Filia McAdams	19653.104
276	MH-17455	Mark Hamilton	19649.456
277	DL-12925	Daniel Lacy	19633.200

278 rows x 3 columns

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SUGGESTED ACTIONS

Build Workspace Show Config

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Describe what to build next

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SQL_TASK3_PROJECT

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Sales_Analysis_Project.ipynb

df.to_sql()

GenerateCodeMarkdownRun AllRestartClear All OutputsPython 3.14.2

5.)What is the highest order value per customer?

```
pd.read_sql("""
SELECT c.customer_id, c.customer_name, MAX(o.Sales) AS highest_order_value
FROM customers c
JOIN orders o
  ON c.customer_id= o.customer_id
GROUP BY c.customer_id, c.customer_name
ORDER BY highest_order_value DESC;
""", conn)
```

[64] ✓ 0.1s Python

	customer_id	customer_name	highest_order_value
0	SM-20320	Sean Miller	22638.480
1	TC-20980	Tamara Chand	17499.950
2	RB-19360	Raymond Buch	13999.960
3	TA-21385	Tom Ashbrook	11199.968
4	HL-15040	Hunter Lopez	10499.970
...	...	...	...
788	CJ-11875	Carl Jackson	16.520
789	MG-18205	Mitch Gastineau	12.320
790	RS-19870	Roy Skaria	9.648
791	LD-16855	Lela Donovan	5.304
792	TS-21085	Thais Sissman	2.808

793 rows x 3 columns

Mini Project: Customer Sales Insights

1. Top 5 Customers

Calculated total sales for each customer and identified the top five revenue-generating customers.

Business Insight: A small group of customers contributes a significant portion of total sales.

2. Bottom 5 Customers

Identified customers with the lowest total sales

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SQL_TASK3_PROJECT

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Mini Project: Customer Sales Insights

1. Top 5 Customers

Calculated total sales for each customer and identified the top five revenue-generating customers.

Business Insight: A small group of customers contributes a significant portion of total sales.

2. Bottom 5 Customers

Identified customers with the lowest total sales.

Business Insight: These customers may require engagement strategies to increase purchases.

3. Customers With Only One Order

Identified customers who placed exactly one order.

Business Insight: These customers represent potential retention opportunities.

4. Customers With Above-Average Sales

Identified customers whose sales exceed the overall customer sales average.

Business Insight: These customers can be targeted for premium offers and loyalty programs.

5. Highest Order Value Per Customer

Calculated the maximum order value placed by each customer.

Business Insight: Highlights each customer's purchasing potential.

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SUGGESTED ACTIONS

Build WorkspaceShow Config

Sales_Analysis_F

Describe what to t

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